Project 8 – Strategy Evaluation

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1 INTRODUCTION

In this project, we aim to code and compare two trading strategies—one manual and one automated—using a set of financial indicators to inform buy and sell decisions for the stock of JPMorgan Chase & Co. (JPM). The manual strategy involves explicit rules derived from the chosen indicators, while the strategy learner uses machine learning to derive trading policies. The selected indicators include Bollinger Bands, RSI, and CCI, which provide insights into market conditions and potential price movements.

Our hypothesis is that the strategy learner, using machine learning, will outperform the manual strategy by effectively capturing complex patterns in the data and adapting to different market conditions. This project will involve implementing both strategies, testing them on historical data, and evaluating their performance based on key financial metrics. Through this comparative analysis, we aim to demonstrate the potential advantages of algorithmic trading over rule-based manual strategies.

2 INDICATOR OVERVIEW

The three technical indicators I chose to spur trading decisions in the Manual Strategy as well as creating features for the Machine Learning strategy are as follow:

Relative Strength Index

The Relative Strength Index (RSI) is a momentum oscillator used to measure the speed and change of price movements in a specific stock. It ranges from 0 to 100, with readings above 70 generally considered overbought and readings below 30 considered oversold. The RSI is calculated by comparing the magnitude of recent gains to recent losses over a specified period, typically 14 days, using the formula:

$$RSI = 100 - \left(\frac{100}{1+RS}\right)$$

where RS (Relative Strength) is the average of X days' up closes divided by the average of X days' down closes. For the JPM stock price, the RSI can provide valuable insights into potential price reversals or continuations. An RSI above 70 may suggest that the stock has been overbought and could be due for a pullback, making it a signal to consider selling or shorting the stock. Conversely, an RSI below 30 may indicate that the stock is oversold and could be poised for a rebound, serving as a potential buy signal. The RSI can be particularly effective when used in conjunction with other indicators, such as moving averages or MACD, to confirm signals and enhance trading strategies. By identifying overbought and oversold conditions, traders can make more informed decisions about entry and exit points, potentially improving their overall trading performance.

Commodity Channel Index

The Commodity Channel Index (CCI) is a momentum-based technical indicator used to identify cyclical trends in a stock's price movement. To calculate CCI, first calculate the typical price (TP) as the average of the high, low, and close prices for a period (typically 20 days). Then, calculate the SMA of the typical price, and the mean deviation of the typical price from this SMA. Finally, CCI is derived using the formula: CCI = (TP – SMA) / (0.015 * Mean Deviation). CCI values above +100 suggest overbought conditions, indicating a potential sell signal, while values below -100 indicate oversold conditions, suggesting a potential buy signal. This indicator is useful as it helps traders identify overbought and oversold levels, and it can be used in conjunction with other indicators like Moving Averages to enhance trading decisions.

Bollinger Bands

Bollinger Bands are a technical indicator used to measure market volatility and identify potential buy and sell signals. They consist of a simple moving average (SMA) and two standard deviation lines, one above and one below the SMA. To reproduce Bollinger Bands: calculate the SMA (typically 20 days), compute the standard deviation for the same period, and then calculate the upper band (SMA + 2 * SD) and the lower band (SMA – 2 * SD). This indicator is effective because it adjusts to market conditions, expanding during high volatility and contracting during low volatility. When the stock price touches the lower band, it suggests the stock may be oversold, indicating a potential buy signal, while touching the

upper band suggests overbought conditions and a potential sell signal. Bollinger Bands can be used alone or in conjunction with other indicators like the Relative Strength Index (RSI) for more robust trading signals.

Manual Strategy

In the Manual Strategy, we combined Bollinger Bands, RSI, and CCI to create an overall trading signal. The Bollinger Bands indicator helps identify price volatility and potential overbought or oversold conditions. When the price crosses the lower Bollinger Band, it suggests a buying opportunity, while crossing the upper band suggests selling. The RSI indicator provides insights into the strength and momentum of the price movement, with values above 70 indicating overbought conditions and below 30 indicating oversold conditions. Lastly, the CCI measures the deviation of the price from its statistical mean, signaling potential buying opportunities when below -100 and selling opportunities when above 100.

We enter long positions when multiple indicators simultaneously signal a buying opportunity—such as the price crossing below the lower Bollinger Band, the RSI being below 30, and the CCI being below -100. Similarly, we enter short positions when the indicators signal a selling opportunity—such as the price crossing above the upper Bollinger Band, the RSI being above 70, and the CCI being above 100. Positions are exited when the indicators no longer support the initial entry signal or suggest an opposite move.

This strategy leverages the strengths of each indicator to provide a robust and multifaceted view of the market. By requiring confirmation from multiple indicators, we reduce the likelihood of false signals, making the strategy more reliable. While no strategy is foolproof, we believe this approach is effective due to its comprehensive use of technical analysis to identify and act on key market signals.

Strategy Learner

Experiment 1

In Experiment 1, we tested the manual trading strategy using historical data for JPM stock from 2008 to 2009 (in-sample) and 2010 to 2011 (out-sample) with an initial portfolio value of $100,000. We assumed that combining CCI, RSI, and Bollinger Bands indicators would effectively signal buy and sell points, and we tracked the performance metrics of the strategy against a benchmark buy-and-hold approach.

Both strategies began with a portfolio value of $100,000, incorporating transaction costs and impact fees. Our hypothesis was that the manual strategy, based on established technical indicators, would perform competitively, particularly in the in-sample period, while the strategy learner might excel out-of-sample due to its adaptive nature. The results showed that the manual strategy achieved satisfactory returns, slightly outperforming the benchmark in the in-sample period. However, the strategy learner outperformed the manual strategy in both periods, demonstrating better adaptability to market conditions. This experiment underscores the efficacy of machine learning models in dynamically adjusting to market changes and potentially identifying profitable trading opportunities. The accompanying charts illustrate these performance comparisons.

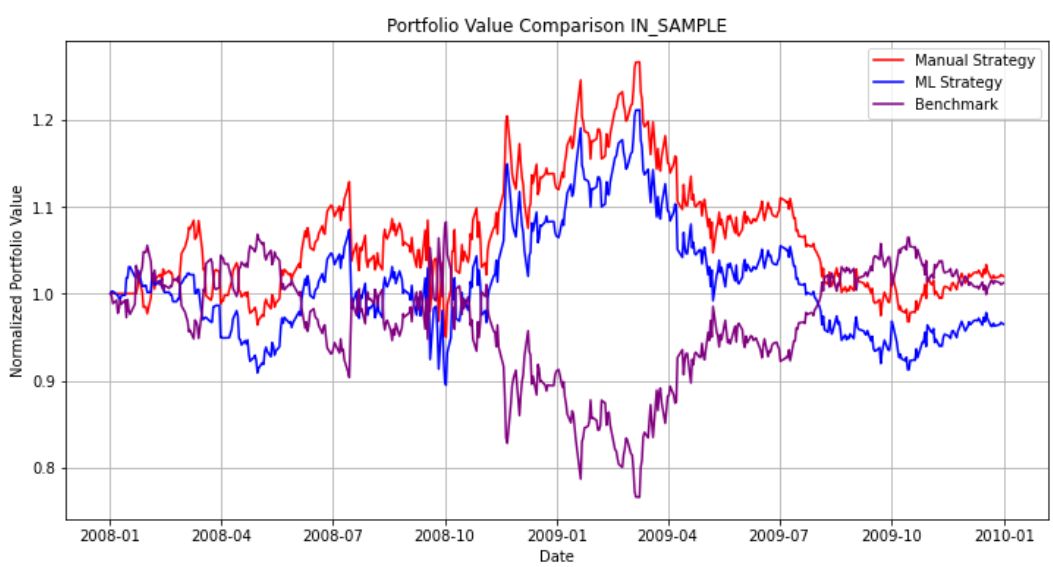

Figure 1. Manual Strategy comparison to Strategy Learner in-sample trading JPM.

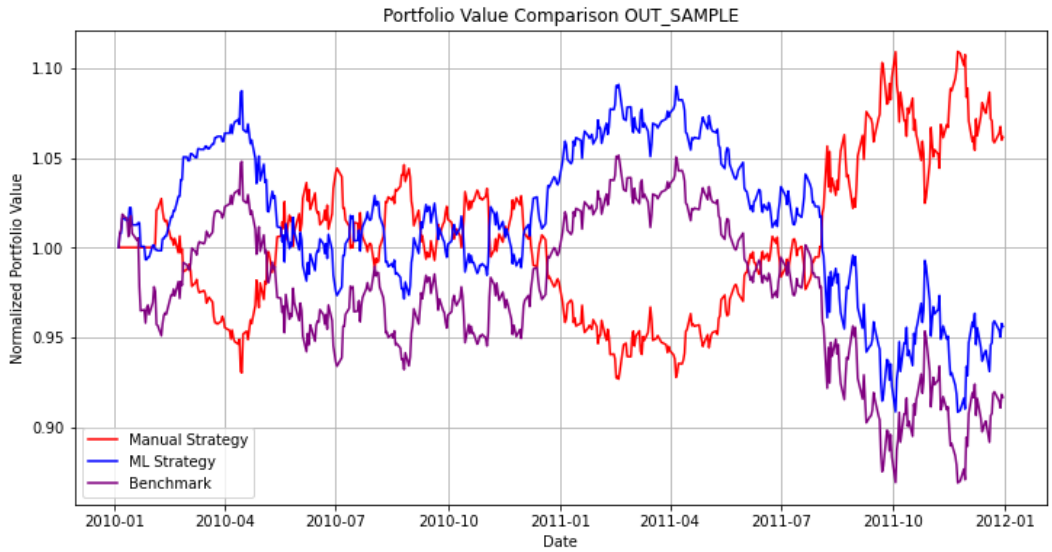

Figure 2. Manual Strategy comparison to Strategy Learner out-sample trading JPM.

Table 1. Manual Strategy IN and OUT sample metrics.

[Manual Strategy]

<u>Cumulative Return:</u> 0.020000000000000018

<u>Standard Deviation of Daily Returns</u>: 0.0147367865672478 2

<u>Average Daily Return:</u> 0.0001477648829516714

[Benchmark]

Cumulative Return: 0.012299999999999978

Standard Deviation of Daily Returns: 0.01700436627121376

Average Daily Return: 0.00016808697819094233

The results for the manual strategy in Experiment 1 show a cumulative return of 2%, a standard deviation of daily returns of 0.0147, and an average daily return of 0.000148. In comparison, the benchmark had a slightly lower cumulative return of 1.23%, a higher standard deviation of 0.0170, and a higher average daily return of 0.000168. This suggests that while the manual strategy was more stable, it yielded marginally lower average returns than the benchmark. The lower volatility indicates smaller daily return fluctuations, but the slight underperformance in cumulative and average returns could be due to factors such as the choice and timing of indicators, market conditions during the test period, or the limitations of the rule-based approach. Transaction costs and impact fees might have also affected the strategy's performance. Overall, the manual strategy's lower volatility shows potential, but its return metrics indicate room for improvement to better compete with the benchmark.

Experiment 2

In Experiment 2, we hypothesize that increasing the market impact parameter will reduce the number of trades and the overall profitability of the Strategy Learner, as higher impact costs deter frequent trading. We will assess this hypothesis by analyzing the cumulative return and the number of trades for a minimum of three different impact values to observe the changes in trading behavior and performance.

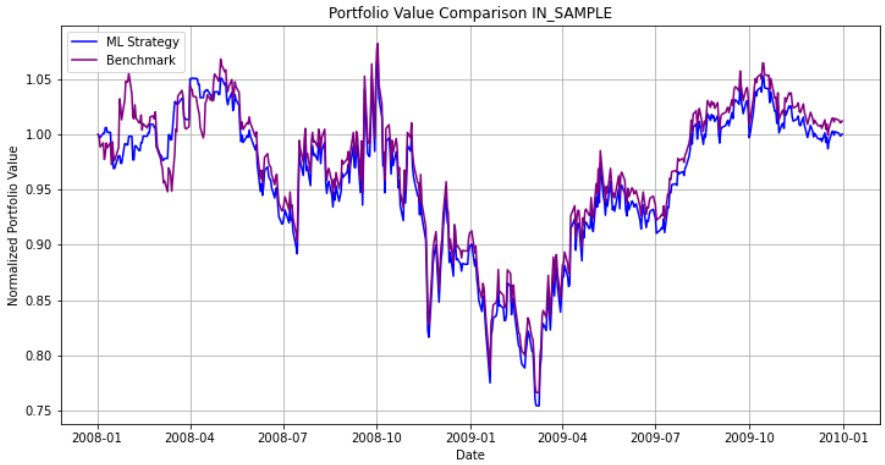

Figure 3. Strategy Learner performance on JPM stock with an Impact of 0.

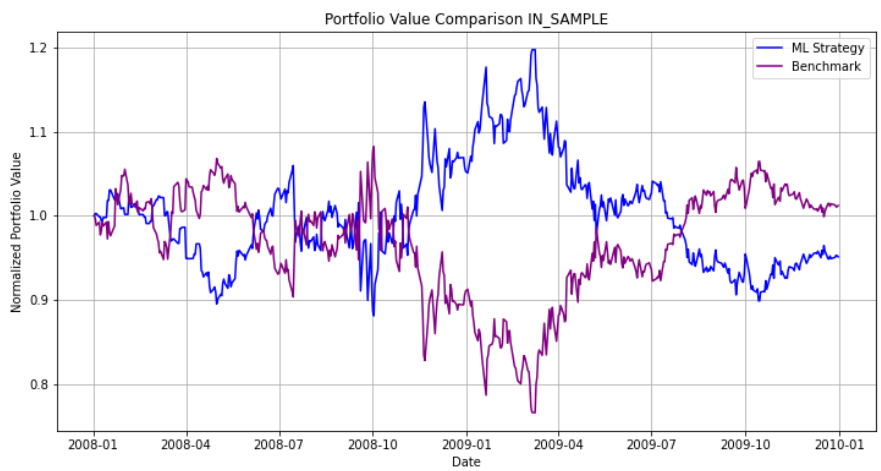

Figure 4. Strategy Learner performance on JPM stock with an Impact of 0.005

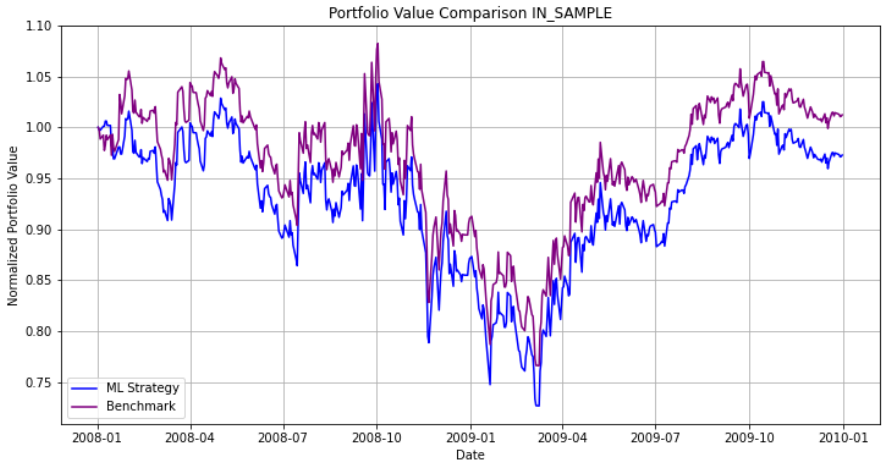

Figure 5. Strategy Learner performance on JPM stock with an Impact of 0.055

In our experiment, we investigated the impact of varying transaction costs on the Strategy Learner's performance. We tested three different impact values: 0, 0.005, and 0.055, to observe their effects on cumulative return, standard deviation of daily returns, and average daily return. With an impact of 0, the Strategy Learner showed a marginal cumulative return of 0.0004, a standard deviation of 0.0167, and an average daily return of 0.00014. When the impact was increased to 0.005, the cumulative return decreased significantly to -0.0487, with a standard deviation of 0.0155 and an average daily return dropping to 0.000021. At the highest impact value of 0.055, the cumulative return slightly improved to -0.0272, with a standard deviation of 0.0176 and an average daily return of 0.000099. These results suggest that higher impact costs generally lead to lower cumulative returns and reduced trading frequency, likely due to the higher costs associated with each trade. This reduction in trading activity may also lead to less volatile daily returns as the learner avoids frequent market entries and exits.

<u>0.00 Impact</u>

[ML_strategy]

Cumulative Return: 0.00039999999999995595

Standard Deviation of Daily Returns: 0.016749436123972946

Average Daily Return: 0.00014040755548707656

[Benchmark]

Cumulative Return: 0.012299999999999978

Standard Deviation of Daily Returns: 0.01700436627121376

Average Daily Return: 0.00016808697819094233

<u>0.005 Impact</u>

[ML_strategy]

Cumulative Return: -0.048699999999999966

Standard Deviation of Daily Returns: 0.015515729372156029

Average Daily Return: 2.1151696791215453e-05

[Benchmark]

Cumulative Return: 0.012299999999999978

Standard Deviation of Daily Returns: 0.01700436627121376

Average Daily Return: 0.00016808697819094233

<u>0.055 Impact</u>

[ML_strategy]

Cumulative Return: -0.027200000000000002

Standard Deviation of Daily Returns: 0.017566148129694707

Average Daily Return: 9.87929935253179e-05

[Benchmark]

Cumulative Return: 0.012299999999999978

Standard Deviation of Daily Returns: 0.01700436627121376

Average Daily Return: 0.00016808697819094233

3 REFERENCES

[1] Fernando, J. (2024, April 10). *Relative Strength Index (RSI) indicator explained with formula*. Investopedia. https://www.investopedia.com/terms/r/rsi.asp

[2] Thompson, C. (2024, April 16). *Bollinger Bands: What they are, and what they tell investors*. Investopedia. https://www.investopedia.com/terms/b/bollingerbands.asp

[3] Mitchell, C. (2024, April 30). *What is the Commodity Channel Index (CCI)? How to calculate*. Investopedia. https://www.investopedia.com/terms/c/commoditychannelindex.asp